



Scratch & Snack - An interactive scratching decorative board for cats and cat owners

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Part 1: Concept and Motivation

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Part 1: Concept and Motivation

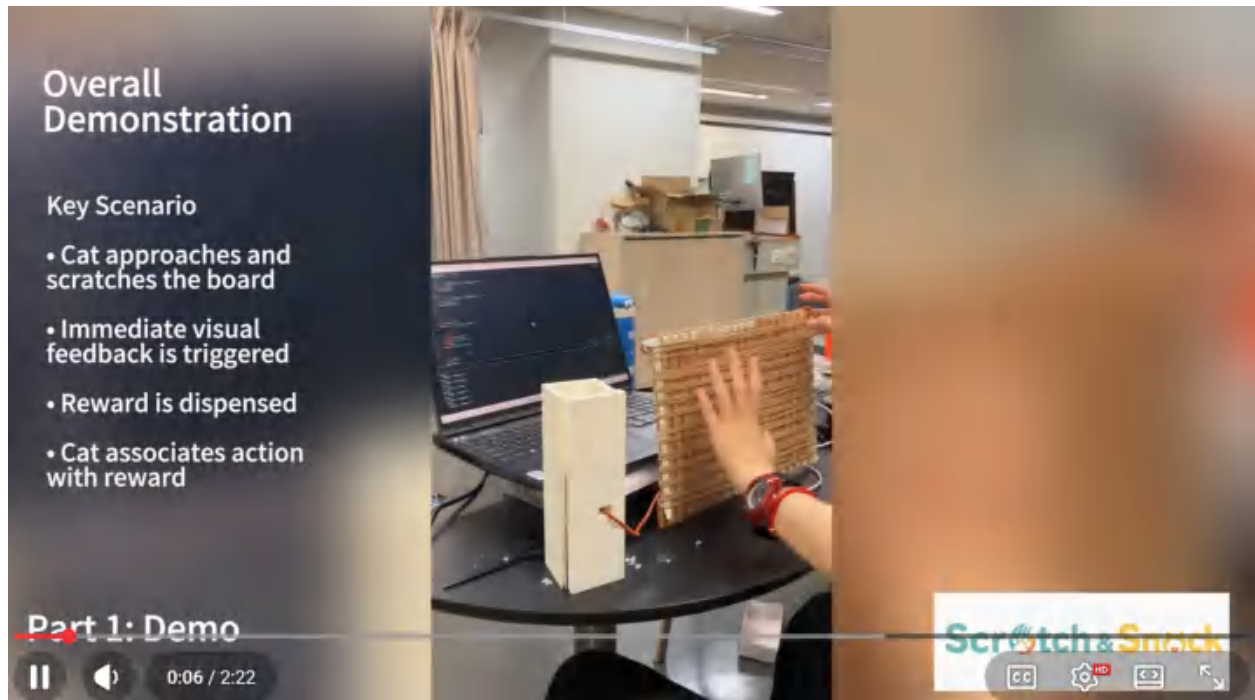
Scratch & Snack is a smart interactive scratching board designed for indoor cats and their owners. Beyond providing feedback and a decorative touch, the LED lights foster interaction between the cat and the owner, transforming scratching from a solitary action into a multi-dimensional experience.

The project originally began as a distributed “Scatter Foraging System” but pivoted to a single, focused object after prototyping revealed the complexity of a multi-node network. The core concept is simple: when a cat scratches the board, a built-in pressure sensor detects the action, triggers an LED feedback, and releases a single treat from a servo-controlled dispenser. This reinforces the cat’s natural scratching behavior through positive reward.

The motivation comes from two user needs. Indoor cats often face limited environmental stimulation and repetitive daily routines, which reduce opportunities for natural behaviors such as scratching. Scratching is instinctive, physically engaging, and linked to territory marking and stress release—making it an ideal trigger for interaction. For owners, traditional toys lose novelty quickly, and uncontrolled scratching damages furniture. **By integrating the toy into a furniture-like form, Scratch & Snack reduces visual clutter, protects household items, and strengthens the pet-owner relationship through a safe, repeatable, and enriching activity. It can be integrated with different scratching materials, put in various places in the house, and modularly assembled in multiple ways.** It aims to enhance the living environment for both the cat and its owners.

Part 2: Video Demo and Assembling Instruction

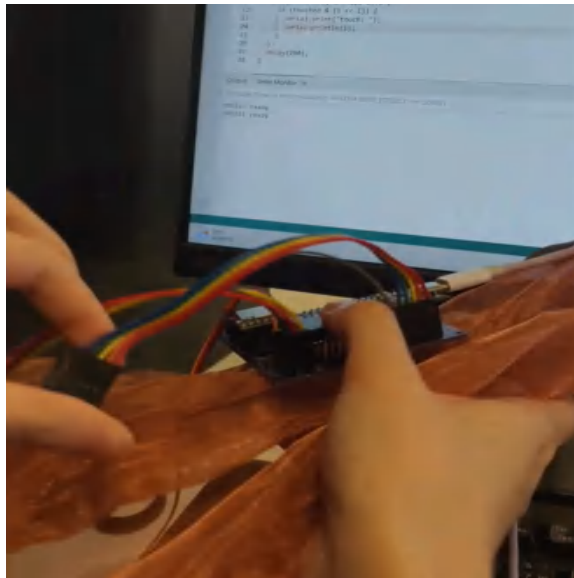
Video Demo



<https://www.youtube.com/watch?v=opdXjLupC50>

Main Modules

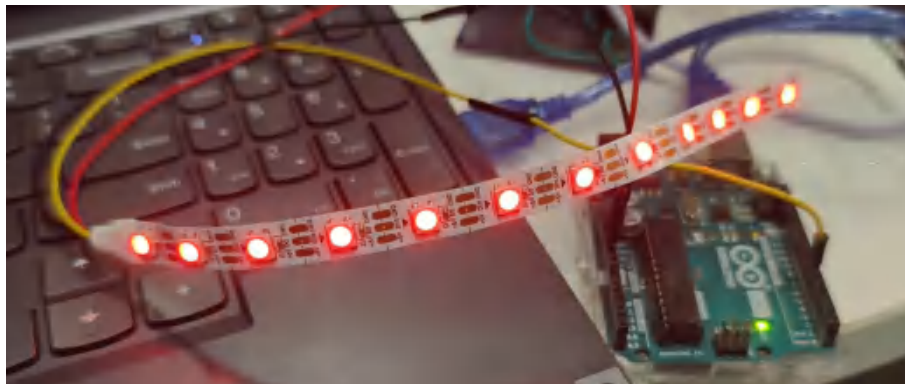
1. MPR121



```
for (uint8_t i = 0; i < 12; i++) {  
  
    // 触網  
    if ((currtouched & _BV(i)) && !(lasttouched & _BV(i))) {  
  
        if (now - lastTriggerTime >= cooldown) {  
  
            lastTriggerTime = now;  
  
            Serial.print("Touched: ");  
            Serial.println(i);  
  
            if (i == 1) {  
                Serial.println(">>> CHANNEL 1 TRIGGERED <<<");  
  
                // 執行雷聲  
                playRedFade();  
                playServoAction();  
            }  
  
            else {  
                Serial.println("Cooldown active");  
            }  
        }  
    }  
  
    lasttouched = currtouched;  
}
```

Solder the copper net to MPR121 channel 1, then touch the net to see if it reacts.

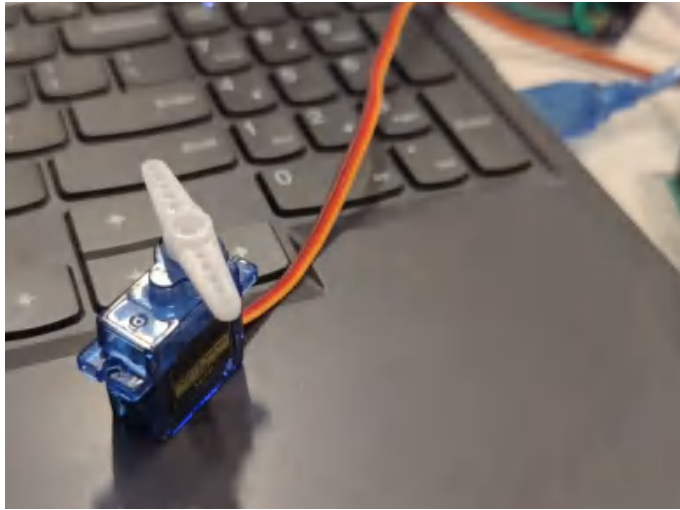
2. LED Light



```
void showColor(uint8_t r, uint8_t g, uint8_t b, float brightness) {  
    for(int i = 0; i < NUMPIXELS; i++) {  
        pixels.setPixelColor(i, pixels.Color((r * brightness), (g * brightness), (b * brightness)));  
    }  
    pixels.show();  
}  
  
void fadeIn(uint8_t r, uint8_t g, uint8_t b, int duration_ms) {  
    if (duration_ms == 0) {  
        showColor(r, g, b, 1.0);  
        return;  
    }  
  
    int steps = 50;  
    int delayTime = duration_ms / steps;  
  
    for(int i = 0; i <= steps; i++) {  
        float brightness = (float)i / steps;  
        showColor(r, g, b, brightness);  
        delay(delayTime);  
    }  
}  
  
void fadeOut(uint8_t r, uint8_t g, uint8_t b, int duration_ms) {  
    if (duration_ms == 0) {  
        showColor(r, g, b, 1.0);  
        return;  
    }  
  
    int steps = 50;  
    int delayTime = duration_ms / steps;  
  
    for(int i = steps; i >= 0; i--) {  
        float brightness = (float)i / steps;  
        showColor(r, g, b, brightness);  
        delay(delayTime);  
    }  
}  
  
// LED 雷聲  
void playRedFade() {  
    fadeIn(255, 112, 139, 2000);  
    delay(2000);  
    fadeOut(255, 112, 139, 800);  
}
```

Set up the color changes of the LED strip.

3. Servo



```
void moveServo(int startAngle, int endAngle, int duration_ms) {
  int steps = abs(endAngle - startAngle);
  if (steps == 0) return;

  int delayTime = duration_ms / steps;

  if (startAngle < endAngle) {
    for (int a = startAngle; a <= endAngle; a++) {
      myServo.write(a);
      delay(delayTime);
    }
  } else {
    for (int a = startAngle; a >= endAngle; a--) {
      myServo.write(a);
      delay(delayTime);
    }
  }
}

// void setup
void playServoAction() {
  moveServo(90, 140, 500);
  delay(300);
  moveServo(140, 90, 500);
}
```

Tune the servo's angle and rotation speed.

4. Button

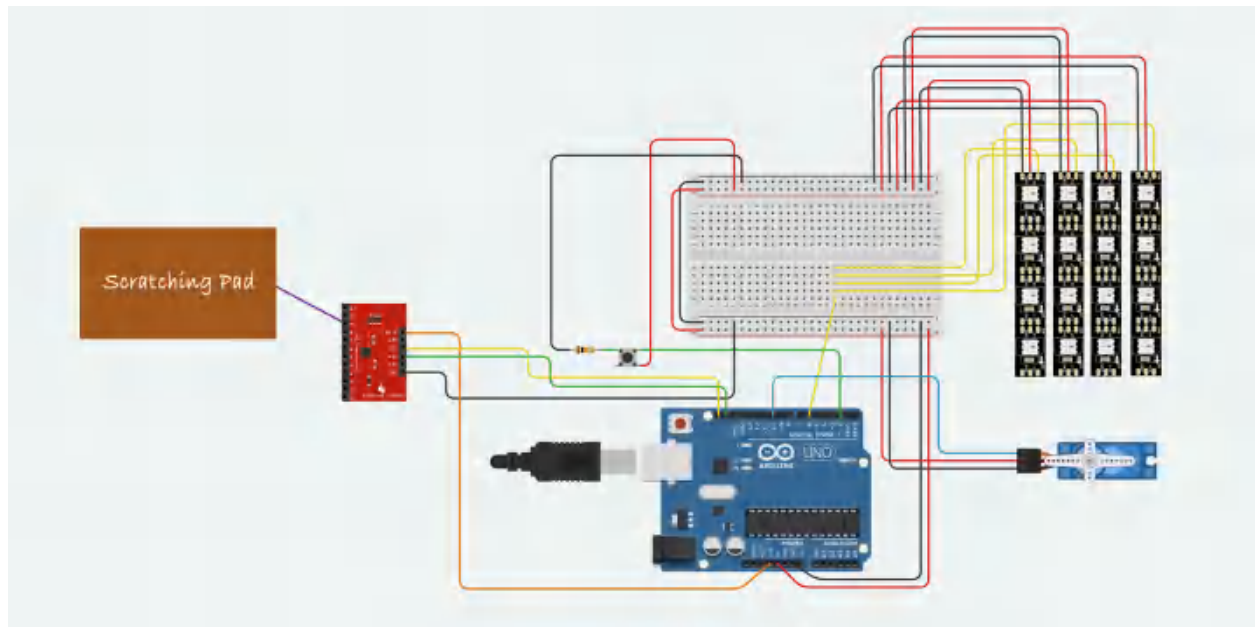


```
if (digitalRead(BUTTON_PIN) == HIGH) {
  void (*resetFunc)(void) = 0;
  resetFunc(); // 软件重启
}
```

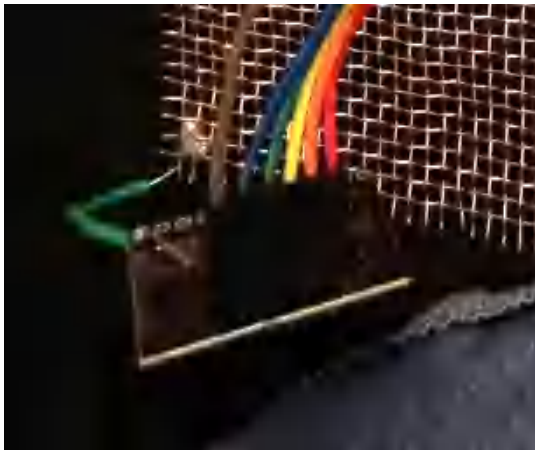
Enable program restart control in the event of a hang or freeze.

Assembly Process

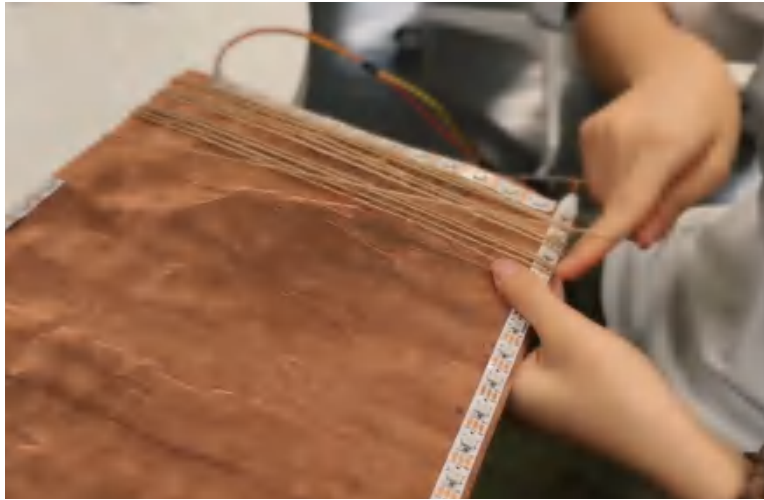
Arduino Part



1.Solder the copper net to channel 1 of the MPR121;



2. Secure the copper net to a rigid board and wrap it with sisal rope of the type commonly used for cat scratching posts;



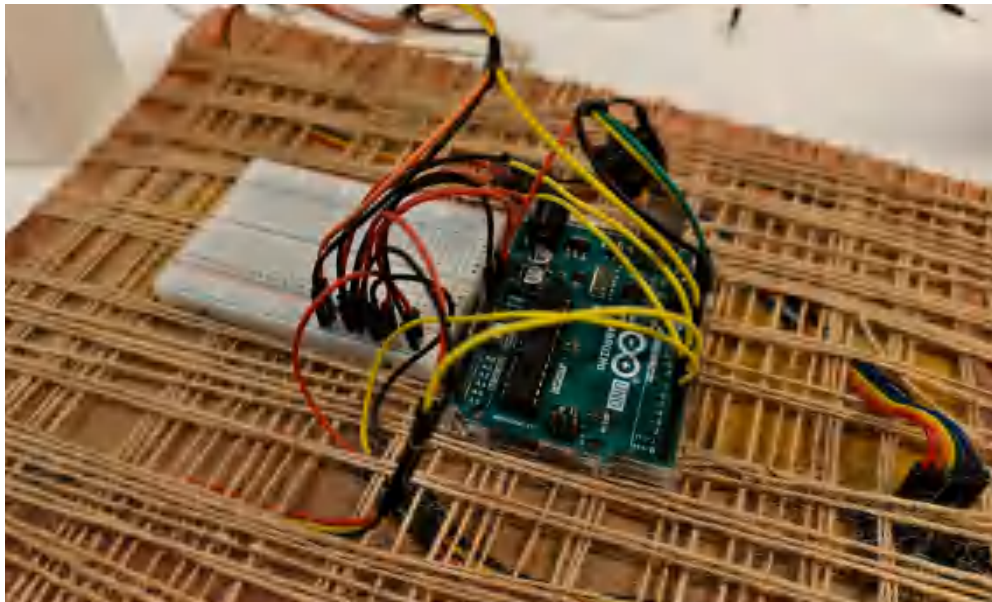
3. Connect the 3.3V, GND, SCL, and SDA pins of the MPR121 to the corresponding pins on the Arduino Uno;

4. Connect the three pins of the LED strip to the 5V, GND, and digital pin 6 on the Arduino Uno. In this setup, the four LED strips are controlled in series.

5. Connect the three pins of the servo to the 5V, GND, and digital pin 2 on the Arduino Uno;

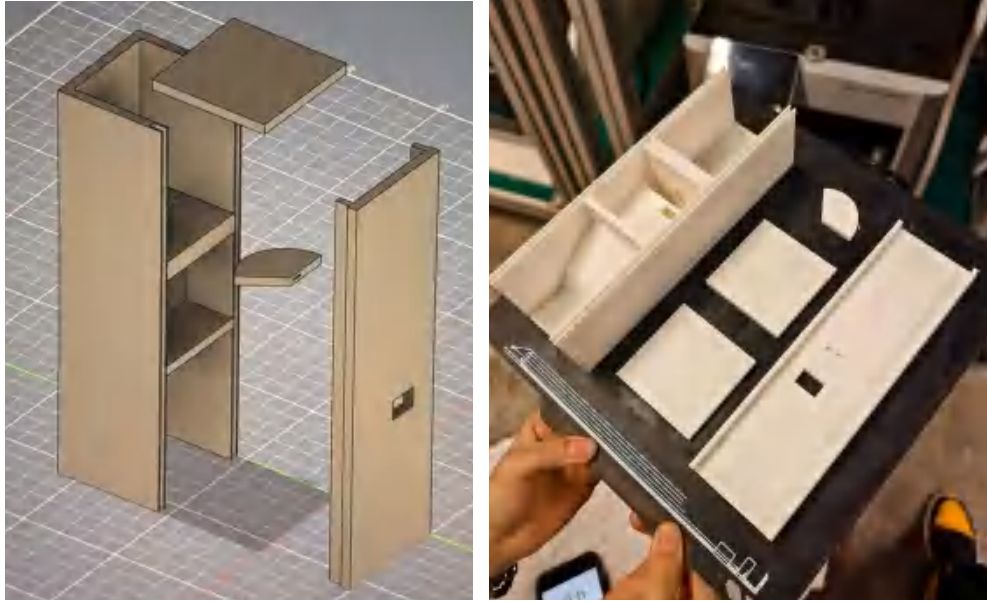
6. Add a button, then connect its three pins to the 5V, GND, and digital pin 2 of the Arduino Uno;

7. Fix all the elements onto the back of the pad.



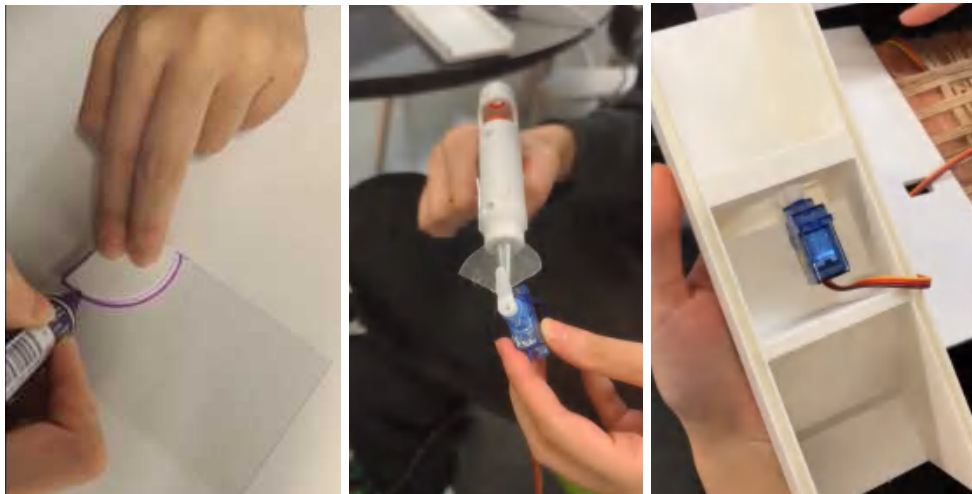
3D Model Part

1. Print the model in four separate parts;

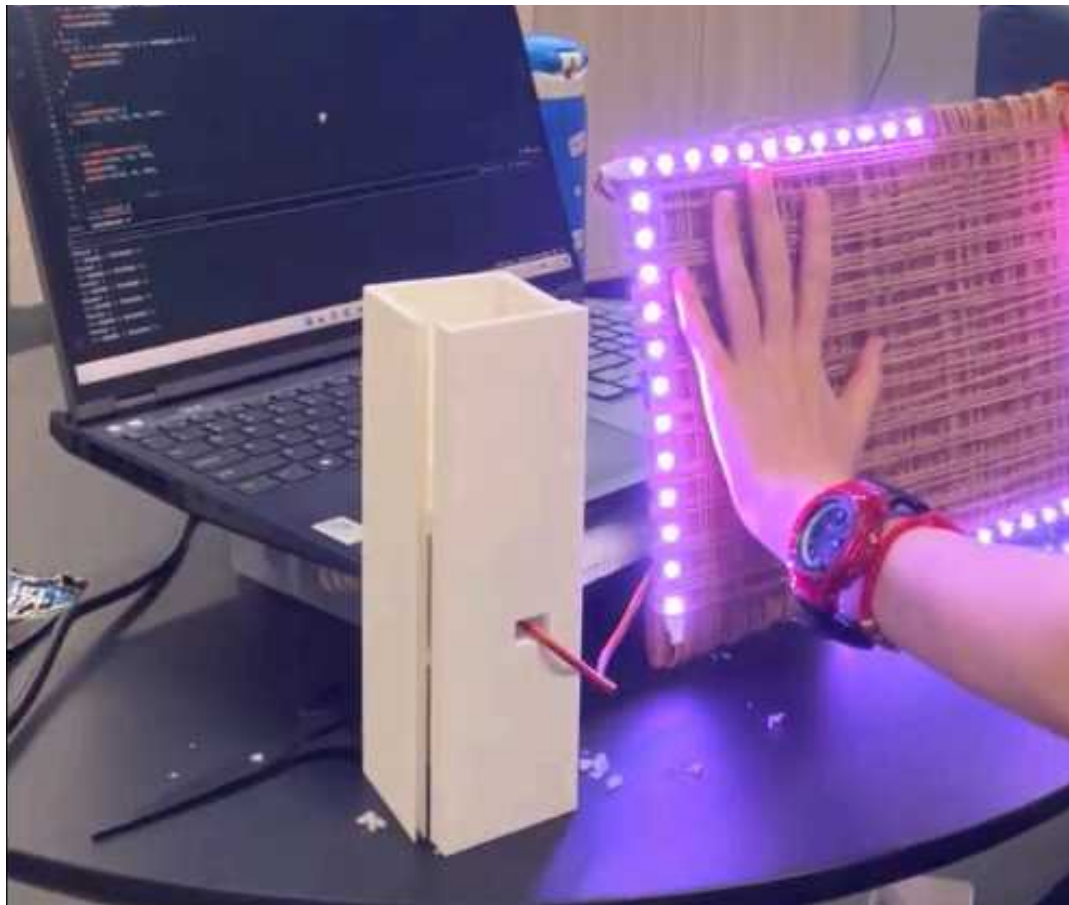


2. Test and fix the servo and the stopper in place.

(Honestly, the stopper we printed came out in the wrong size, so we ended up making a custom one ourselves (^))



Assemble Everything



Part 3: Source files (3D model and Arduino code)

 Ginandtoshi Add files via upload	ae27ca9 · now	🕒 12 Commits
 Scratch-Snack	Arduino	4 hours ago
 LICENSE	Initial commit	5 hours ago
 README.md	Update README to use image syntax for toy final image	8 minutes ago
 Toy final ppt.pdf	Add files via upload	now
 toy final product overview.png	Add files via upload	12 minutes ago
 toy.3mf	Add files via upload	5 hours ago
 toy.gcode.3mf	Add files via upload	5 hours ago

Access through GitHub: <https://github.com/Ginandtoshi/Scratch-Snack>

Part 4: Component list

Name	Quantity	Link to purchase
麻绳diy 1.5mm x 2m	1	【淘宝】 https://e.tb.cn/h.iwtgbt3ETkJCnHq?tk=6Gf15NZMF2A CZ009 「麻绳麻绳diy优质手工绳子天然粗细麻绳工艺装饰品吊牌捆绑绳包邮」 点击链接直接打开 或者 淘宝搜索直接打开
2812 LED light strip arduino 5V 20cm 60	4	【淘宝】假一赔四 https://e.tb.cn/h.iwtgiytp5jKNTub?tk=WDsw5NZMyIX MF287 「2812语音灯带DIY制作套件arduino编程大学生单片机5V灯条 单点单控」 点击链接直接打开 或者 淘宝搜索直接打开
猫抓板猫垫子 40cm x 60cm	1	【淘宝】7天无理由退货 https://e.tb.cn/h.iwtgRopfa0HFUdU?tk=g9Rj5NZo5FT CZ356 「猫抓板 猫抓垫猫地毯仿剑麻宠物地垫防滑猫咪蹭痒垫睡垫猫狗垫大码」 点击链接直接打开 或者 淘宝搜索直接打开
Arduino Uno	1	【淘宝】假一赔四 https://e.tb.cn/h.ixVOWXBsTq5TEsf?tk=YWbR5NZJkx MF287 「意大利 原装 arduino uno r3 学习开发板套件 scratch mixly编程」 点击链接直接打开 或者 淘宝搜索直接打开
紫铜网 80目 30cm x40cm	1	https://mobile.yangkeduo.com/goods2.html?ps=ouWtxP2gdH